

TOPICAL REVIEW

Content Advisors: Lee H. Schwamm, MD, and Gisele S. Silva, MD, MPH, PhD

Balancing Innovation and Responsibility: Ethical and Privacy Challenges in Stroke Digital Health

Marian Muchada¹ MD, PhD*; Federica Rizzo² MD*; Nicoletta Brunelli³ MD, PhD; Carlos A. Molina⁴ MD, PhD

ABSTRACT: Digital health solutions are being increasingly used to support care delivery across various medical fields. In stroke care, technology has contributed to enhancing prehospital workflows, facilitating communication among health care professionals, and enabling fast-track systems for hyperacute stroke management. Furthermore, digital tools are used to improve etiological diagnosis and allow for more tailored long-term follow-up of stroke patient survivors. Digital technologies have become an integral part of stroke management, and the rapid evolution of artificial intelligence continues to accelerate and drive this transformation. However, ethical and privacy considerations are central and represent a major challenge for the design and implementation of digital solutions in health care. Health care professionals involved in stroke care require a clear understanding of the digital tools they use, including their potential, limitations, and biases, and must be aware of the responsibilities inherent in relying on such systems for treatment decisions and outcome prediction. This topical review aims to examine current digital innovations in stroke care, with a particular focus on the key ethical and privacy challenges they raise.

Key Words: artificial intelligence ■ digital health ■ health care ethics ■ health information technology ■ information processing ■ privacy ■ stroke care

Stroke digital technologies encompass all technological solutions aimed at improving patient care at every stage of the stroke care pathway. The underlying concept of a digital solution is to support clinicians in their decision-making while ensuring widespread access to stroke care, even in areas where geographic or geopolitical barriers may hinder timely treatment. The integration of digital technologies into stroke care offers significant potential to enhance patient outcomes and streamline health care delivery. However, these advancements also introduce critical ethical and privacy challenges that must be addressed to ensure their equitable and responsible implementation. One primary concern is the risk of exacerbating existing health disparities. Vulnerable populations, including the elderly, individuals with low socioeconomic status, and those residing in rural or underserved areas, or in low-middle-income countries, often face barriers to accessing digital health solutions due to limited digital literacy, inadequate infrastructure, or financial constraints.^{1,2}

Moreover, the design and deployment of digital health tools must consider ethical principles such as inclusivity, transparency, and accountability. Ensuring that these technologies are developed with input from diverse populations and are adaptable to various cultural and socioeconomic contexts is essential for their effectiveness and acceptance.³ Privacy concerns are also paramount. The collection and use of personal health data through digital platforms raise issues related to data security, informed consent, and potential misuse of information. Patients may be unaware of how their data is collected, stored, and shared, leading to potential breaches of confidentiality and trust.⁴ Although digital health innovations offer significant potential for enhancing stroke care, it is imperative to address the associated ethical and privacy challenges in a proactive and systematic manner. Ensuring that these technologies are developed and implemented with consideration for equity, transparency, and data protection is essential to fostering a health care system that is inclusive, trustworthy, and responsive to the

Correspondence to: Marian Muchada, MD, PhD, Stroke Unit, Department of Neurology, Vall d'Hebron University Hospital, Vall d'Hebron Research Institute, Departament de Medicina, Universitat Autònoma de Barcelona, Ps. Vall d'Hebron, 119-129, 08035, Barcelona, Spain. Email muchada.marian@gmail.com

*M. Muchada and F. Rizzo contributed equally.

For Sources of Funding and Disclosures, see page 1083.

© 2025 American Heart Association, Inc.

Stroke is available at www.ahajournals.org/journal/str

needs of all individuals, regardless of socioeconomic status, geographic location, or demographic characteristics.

This article is structured as a narrative review, aiming to synthesize key themes, challenges, and developments across the literature on artificial intelligence (AI) digital health in patients with stroke. A nonsystematic, exploratory search of PubMed and Google Scholar was conducted, employing search terms including “digital health,” “artificial intelligence,” “stroke,” “ethics,” and “privacy”. Priority was given to recent reviews, articles, and seminal studies.

DIGITAL SOLUTIONS IN STROKE CARE

Prehospital Phase: Digital Triage Systems, Emergency Alert Apps, Geolocation Technologies, Telestroke System

Emergency Medical Services play a critical role in acute stroke care by enabling early evaluation and notification, expediting in-hospital response, and improving patient outcomes.

Digital triage systems and validated tools for large vessel occlusion detection enhance diagnostic accuracy and support direct transport to comprehensive stroke centers.^{5,6} Mobile stroke units, with onboard imaging and telecommunication systems, reduce treatment times and improve outcomes, though high costs and logistical complexity limit widespread use.⁷ Ambulance-based telemedicine offers a scalable, cost-effective alternative with high diagnostic accuracy and remote neurological assessment.⁸

Despite these advances, delayed recognition of stroke symptoms by patients or caregivers remains a major barrier to timely emergency medical services. AI-powered tools are emerging to address this gap.^{9,10} Wearables with motion sensors detect gait and motor abnormalities, while AI-driven smartphone and computer applications analyze speech, language, and motor patterns to identify early neurological changes, achieving high accuracy in detecting neurological signs.^{9,11} AI-based eye-tracking systems can detect gaze deviation, a reliable marker of large vessel occlusion.¹² Integrating automated recognition of key deficits into unified digital platforms may further improve early stroke detection and prehospital triage.¹²

AI also enhances emergency medical services operations by improving dispatcher performance and optimizing resource allocation. Automated speech recognition can increase stroke detection by 16% and thrombolysis use by 2%,¹³ while intelligent algorithms for ambulance deployment improve response efficiency and dynamically allocate resources based on regional and temporal demands.^{14,15}

Digital health platforms facilitate real-time collaboration among stroke specialists via telestroke systems and

prearrival notification apps, supporting rapid multidisciplinary decision-making. Seamless integration of pre-hospital data into hospital information systems ensures faster, coordinated care and higher reperfusion treatment rates.¹⁶

In-Hospital Management: Decision Support Tools, AI-Assisted Imaging, Interdisciplinary Communication Systems

Recent advancements in AI and machine learning are changing stroke diagnosis, especially in the acute phase, where time is crucial. One prominent area of development is AI-powered imaging analysis, where deep learning models, such as convolutional neural networks, are used to detect infarcts, classify hemorrhages, identify stroke due to large vessel occlusion, and volume of irreversibly damaged or potentially salvageable tissue.^{17,18} These models have demonstrated performance equal to or exceeding human radiologists, and several US Food and Drug Administration-approved tools are now available in clinical settings to assist with the rapid interpretation of computed tomography and magnetic resonance imaging scans.¹⁶ This streamlining of diagnostic workflows, processed directly at the scanner, can alert the stroke team immediately, reducing the time to initiate an acute stroke consultation. It allows for faster triage and treatment, thereby improving outcomes in time-sensitive scenarios, such as ischemic strokes.¹⁹ Decision-support platforms integrate medical data, imaging analysis, and evidence-based guidelines to facilitate timely decisions appropriate for thrombolytic therapy for ischemic strokes or interventions for hemorrhagic strokes. Furthermore, large language models like ChatGLM-6B are being explored for their ability to facilitate stroke diagnosis and treatment from electronic health records in addition to noncontrast computed tomography reports, demonstrating the growing synergy between natural language processing and clinical decision support.²⁰

Posthospital and Rehabilitation Phase: Remote Monitoring, Wearable Devices and Sensors, Apps for Cognitive or Physical Rehabilitation Exercises

Mobile health technologies such as smartphone applications, wearables, and sensors offer opportunities for remote monitoring of patients' vital signs, medication adherence, and rehabilitation progress, enabling personalized and proactive management of patients with stroke.

Home-based monitoring tools, such as wearable ECG and photoplethysmography systems, facilitate early detection of atrial fibrillation (AF), enhancing stroke prevention treatment. Prolonged monitoring has been shown to improve the detection of occult AF in patients with

ischemic stroke.²¹ Although implantable cardiac monitors offer the most definitive solution, they have shown the ability to detect AF in 12.4% and 30% of patients at 12 and 36 months, respectively.^{21,22}

These findings are consistent with the results of the STROKE AF trial (The Stroke of Known Cause and Underlying Atrial Fibrillation),²³ which involved patients with suspected noncardioembolic stroke, and the CRYSTAL AF trial (The Cryptogenic Stroke and Underlying AF) cryptogenic stroke.²² Noninvasive monitoring methods, such as 30-day external loop recorders, have also demonstrated substantial effectiveness, while being less costly and less invasive.²⁴ Additionally, consumer-grade smartwatches are emerging as a promising, noninvasive, and cost-effective alternative. These devices passively collect heart rate data and detect irregular rhythms using photoplethysmography technology. Although less accurate than ECG, photoplethysmography-based algorithms have shown strong performance in AF detection, with some studies reporting a positive predictive value of up to 98%.²⁵

Digital health solutions, such as stroke-specific mobile apps and virtual reality therapy, support home-based rehabilitation by offering guided exercises, progress tracking, and personalized feedback. These technologies enhance patient engagement, therapy intensity, and neuroplasticity, leading to improved functional and cognitive recovery. A randomized controlled trial found that a virtual reality-based telerehabilitation program significantly improved upper limb motor function in patients with stroke.²⁶ Another study, exploring the effectiveness of a mobile app-based telerehabilitation program for patients with poststroke, showed significant improvements in motor function, balance, and quality of life compared with traditional in-person therapy.²⁷ This system, demonstrating high accuracy in motor recovery assessment and cognitive evaluation, could offer a comprehensive solution for continuous and personalized care in a home setting.

Patients With Stroke Follow-Up and Secondary Prevention: Platforms for Personalized Follow-Up, Prediction of Stroke Recurrence Risk

Digital health can support secondary stroke prevention by enabling continuous remote monitoring, improving clinical decision-making, and facilitating the delivery of personalized, patient-centred care.²⁸ For instance, programs delivered using digital health have been effectively used to improve cardiovascular risk factor control, medical adherence, and patient outcomes.^{29,30} A recent pilot study has evaluated the Care Assistant and Support Program, a novel multicomponent digital health intervention designed to support secondary prevention in patients with stroke and transient ischemic attack.³¹ This study consisted of a 12-week program that integrated a mobile application, wearable activity tracker, clinician portal, and SMS-based communication to promote

self-management and health behavior change. Usability was evaluated as good to excellent, and participants reported high satisfaction, particularly in terms of self-monitoring and goal engagement, such as improvements noted in self-efficacy, cardiovascular health, and mental health.³¹ Another significant area of development of AI is the prediction of stroke outcome. Engaging and developing deep learning techniques can help to analyze large and different medical data, including brain scans, medical reports, and other sensor information.³² In this context, an emerging theme is the use of real-time physiological monitoring data for predictive modeling. Studies have developed machine learning techniques that process real-time hemodynamic signals to diagnose stroke subtype and predict mortality, stroke recurrence, and treatment outcomes within minutes to hours.³³ These tools are particularly valuable in settings with limited imaging availability, offering a noninvasive, continuous means of risk assessment. Furthermore, deep learning-based models can predict functional outcome at 90 days after thrombectomy using clinical variables and magnetic resonance imaging shortly after treatment with reduced subjectivity and user burden.³⁴

To date, significant progress has been made in the prediction of stroke outcome. By using multimodal data and advanced model architectures, some potential avenues for future research could include: the development of adaptive multimodal fusion methods that effectively combine complementary data sources, the consideration of lesion evolution across time, the adoption of federated learning (FL) techniques for enhanced data utilization, and the exploration of annotation-free lesion labeling methods.³² Early identification of patients at high risk of stroke recurrence, to enable personalized intervention strategies, is a key goal for stroke physicians. The PRERISK model (A Personalized, Artificial Intelligence Based and Statistically Based Stroke Recurrence Predictor for Recurrent Stroke),³⁵ developed from nearly 42000 patients with stroke in Catalonia using clinical and socioeconomic data from the public health care system, combines AI and statistical methods to estimate individualized risk of recurrent stroke. It predicts recurrence risk across 3-time horizons (within 90 days, 91–365 days, and beyond 365 days) and allows for dynamic integration of modifiable risk factor control, and potentially treatment adherence data, supporting its application in clinical practice and precision medicine.

ETHICAL CHALLENGES

Digital Health Equity

Digital health solutions aim to enhance medical care and support clinicians in their work; at the same time, the digitalization of health care raises numerous ethical concerns.

Although digital solutions, such as telestroke systems and integrated networks between emergency

medical services and stroke centers, can improve access to stroke care,¹⁶ their effectiveness relies on adequate infrastructure and sustained investment at the regional and national levels. Without these, implementation may be limited, potentially widening existing disparities in health care access both between countries with different income levels and.³⁶ It is, therefore, essential to consider both the applicability and versatility of any new digital solution from the earliest stages of development.

A critical dimension of digital inequity exists at both the community/institutional and individual patient level.³⁷ At the community level, disparities arise from gaps in infrastructure, funding, and capacity, such as rural hospitals or low-middle-income countries lacking reliable internet, updated tools, or trained staff, limiting access and raising ethical concerns about data governance and patient privacy. At the patient-level, barriers include lack of devices, low digital literacy, socioeconomic challenges, and impairments from stroke, which hinder use of digital health tools and pose privacy risks. Addressing inequities at both levels is essential to ensure that digital health solutions are ethical, effective, and applicable across diverse health care settings, including low- and middle-income countries, thereby supporting equitable stroke care and protecting patient privacy.³⁷

Apart from digital inequities, we must also face inequalities in other areas of everyday life when working towards an equitable health care system. According to the United States Department of Health and Human Services, social determinants of health (SDoH) are the conditions in which people are born, grow, live, work, and age.³⁸ These factors have a profound impact on health outcomes and health equity and are commonly categorized into 5 key domains:

1. Economic stability: employment status, income level, and financial security.
2. Education access and quality: literacy, language skills, educational opportunities.
3. Healthcare access and quality: availability of health services, insurance coverage, quality of care.
4. Neighborhood and built environment: housing quality, transportation, safety, environmental conditions.
5. Social and community context: social integration, support networks, discrimination, community engagement.

While these SDoH shape overall health outcomes, the increasing role of technology in health care and daily life introduces additional factors known as digital determinants of health (DDoH).³

These include:

1. Access to technological tools: availability of devices such as smartphones, computers, and health monitoring tools.
2. Digital literacy: the ability to effectively use digital devices, platforms, and applications to access health information and services.

3. Community digital infrastructure: quality and availability of broadband internet and network connectivity within communities

DDoH acts both independently and synergistically with SDoH. For example, limited access to broadband internet (DDoH) can exacerbate health care access issues (SDoH) by restricting telehealth use or access to online health information. Similarly, poor digital literacy can hinder an individual's ability to engage with digital health tools, negatively impacting disease management and preventive care.³⁹ In this way, DDoHs can serve as both barriers and facilitators of health, influencing the effectiveness of interventions aimed at reducing health disparities. As health care increasingly integrates digital technologies, addressing DDoH alongside traditional SDoH becomes essential for promoting equitable health outcomes.

Fairness of AI Algorithms in Stroke Care

The incorporation of AI into stroke management holds promise for enhancing diagnostic accuracy and treatment personalization. However, the reliance on AI algorithms introduces significant ethical concerns due to biases in data and model development processes. AI systems are often trained on historical medical data, which may reflect existing health care disparities. Consequently, algorithms can perpetuate these biases, leading to inequitable care outcomes. An AI algorithm designed to predict complex care needs exhibited biased results along racial lines, potentially depriving certain patient populations of appropriate care.⁴⁰ At the same time, training algorithms on large, heterogeneous real-world clinical data allows access to more diverse populations than randomized clinical trials, which are often limited by strict inclusion and exclusion criteria.

Algorithms trained on historically biased data can cause disproportionate harm, making the evaluation and mitigation of such harm central to fairness in machine learning.²⁷

However, despite awareness of the sample composition and the data used for training, predictive model results may sometimes lack sufficient discrimination and calibration.⁴¹ Therefore, it is necessary to improve modeling strategies to address disparities in stroke prediction and care.

The generalizability of AI models relies on diverse training data sets that reflect various patient demographics, clinical conditions, imaging modalities, social status, and other relevant factors. Differences in health care site workflows, including clinical protocols, data collection methods, and technological infrastructure, also affect AI performance in new settings. Without diverse data and understanding of site-specific variations, AI tools may perform poorly for underrepresented groups. Biases from unbalanced data and algorithm design can worsen

health care disparities and inequalities.⁴² To reduce group fairness violations, bias-mitigation techniques can adapt existing algorithms through preprocessing (eg, blind-ing, augmenting, or reweighting data), in-processing (eg, adding fairness constraints or regularization terms), and postprocessing (eg, calibrating predictions across subgroups). Teams must also establish postdeployment monitoring of AI algorithm performance and implement updating protocols that incorporate technical solutions and clinical inputs into an AI safety checklist to ensure ongoing safety, fairness, and reliability in real-world clinical settings.⁴³ This monitoring involves continuously evaluating the algorithm's accuracy across diverse patient populations, detecting potential biases or errors that may emerge over time, and assessing its impact on clinical decision-making and patient outcomes. Additionally, this process should incorporate mechanisms for timely updates and recalibration of the algorithm as new data become available, thereby maintaining its effectiveness and minimizing unintended harms or disparities in care.⁴⁴

Biases in health care AI, including data, algorithmic, deployment, and data set shift, can undermine fairness and safety, particularly in stroke care.⁴⁵ Mitigation strategies aim to detect, reduce, or eliminate sources of unfairness that may arise during the development, training, and deployment of AI systems.⁴⁶ These strategies require a lifecycle approach, integrating bias detection, human oversight, and continuous validation to ensure equitable and reliable clinical AI. Herein, we provide a comprehensive overview of the different types of bias and strategies developed to assess and mitigate them (Table).^{19,36,47–51}

Transparency, Clinical Decision-Making, and Accountability

Once different types of bias and the corresponding mitigation strategies are understood, it becomes clear that these considerations are essential for ensuring algorithmic transparency and accountability, 2 foundational principles when integrating AI into clinical decision-making. Transparency allows clinicians to understand how AI systems generate outputs, at least partially. This becomes particularly relevant in clinical contexts where algorithmic recommendations influence diagnosis, prognosis, or treatment choices. However, many AI models, particularly Differential Privacy, are often described as black boxes, limiting interpretability.⁵² Accountability refers to the ability to attribute responsibility for a model's decisions. In clinical contexts, accountability means that when an AI system is implemented, it should be clear who holds responsibility for its recommendations: developers, health care institutions, clinicians, or regulators. At this point, AI can introduce a responsibility gap, where those affected by algorithmic decisions, such as patients harmed by an incorrect recommendation, may not have a clearly identifiable person or entity to hold accountable, particularly

when biases disproportionately affect vulnerable or marginalized populations. The extent of this gap varies by region, with different regulatory laws. In low- and middle-income countries, this gap can be even wider due to weaker regulatory structures. Globally, closing this gap will require regulation, education, and shared accountability to ensure safe and equitable patient care.⁵³ To address these challenges, regulatory and ethical frameworks are essential. International bodies like the World Health Organization have emphasized the need for AI systems that are transparent, accountable, and equitable in health care. Their 2021 guidance outlines 6 principles for the ethical use of AI in health, including promoting human autonomy, ensuring safety and transparency, and fostering inclusiveness and equity.⁵⁴ Additionally, the Framework for Digital Health Equity³ advocates for an upstream, multi-level approach that considers the DDoH. It also stresses that digital tools should be designed to be flexible and scalable, especially for low-resource settings with limited digital infrastructure. In summary, bias identification and mitigation are not merely technical concerns—they are tightly linked to the broader goals of ethical and responsible AI. Transparency and accountability are indispensable to building trust in clinical AI systems, and they must be actively supported through careful model design, stakeholder engagement, and clear regulatory guidance.

PRIVACY AND DATA SECURITY ISSUES

As digital health technologies (DHTs) continue to reshape stroke care, it is critical to address the ethical and privacy challenges they present. Effective management of data collection, storage, and third-party access is essential to ensure that patient privacy is upheld and that patients can trust the systems designed. By adhering to regulatory frameworks like the Health Insurance Portability and Accountability Act and General Data Protection Regulation, and implementing cybersecurity measures, health care providers and technology companies can mitigate risks. Ultimately, this patient-centred approach prioritizes informed consent, data security, and transparency is key to the ethical integration of digital health in stroke care.

Addressing these challenges can foster an ethical and privacy-conscious environment that promotes patient well-being while respecting their rights and personal data.^{55,56}

Data Collection and Consent

DHTs in stroke care, such as mobile apps, wearables, telemedicine, and AI tools, support a shift toward value-based health care focused on improving patient outcomes through standardized practices. These tools collect both medical and nonmedical data to enable more personalized and equitable care. However, they also raise ethical

Table. Overview of Bias Types and Mitigation Strategies in Stroke-Related AI Systems

Type of bias	Definition	Subtypes	Stroke-related example	Mitigation strategies
Data bias	Originates from limitations or imbalances in training data, including how they are collected, labeled, or sampled. Leads to systematic underrepresentation of specific populations.	Sampling bias: nonrepresentative data sets. Measurement bias: errors or inconsistencies in how data are collected or labeled. Sample-selection bias: occurs when the data set used to train a model is not representative of the broader patient population, often reflecting historical inequities.	The <i>Stroke Riskometer</i> ⁴⁷ based on the mainly white, male Framingham cohort, performs poorly in other ethnic groups. Delayed referrals for Black or Hispanic patients lead to underrepresentation of early-stage cases. ⁴⁸	Preprocessing approaches: Detect and reduce bias before training through resampling, reweighting, or data transformation to balance subgroup representation. Blinding techniques can remove proxies for race or sex. In stroke care, oversampling minority or rural cases improves model fairness.
Algorithmic bias	Bias introduced or amplified by model architecture, training procedures, or proxy variables, even when input data are unbiased.	Overfitting to the majority group: The model learns subgroup-specific patterns. Proxy bias: indirect encoding of protected attributes. Feedback loop bias: reinforcement of biased past decisions.	Stroke detection algorithms show high accuracy in White patients but underperform for minorities. ⁴⁷ Use of zip code as a variable encodes socioeconomic disparities. ⁴⁹ AI systems recommend fewer aggressive treatments for certain racial groups because historical data reflect biased care. ⁴⁸	In-processing approaches: modify learning algorithms to enforce fairness. Add nondiscrimination constraints or loss penalties that reduce model reliance on sensitive attributes. Adversarial debiasing trains an auxiliary network to prevent subgroup discrimination, ensuring stroke models perform equally across sex or age.
Deployment bias	Arises during or after deployment, shaped by contextual, organizational, or social factors that affect how AI outputs are interpreted or who benefits from them.	Interpretation and use bias: misinterpretation or overreliance on AI predictions. Access bias: inequitable access to AI-enabled health care tools.	Clinicians overly trust AI-generated stroke severity scores, ignoring social or behavioral determinants. ⁴⁵ Telemedicine stroke monitoring systems remain inaccessible in rural or low-income areas. ³⁶	Postprocessing and governance strategies: Maintain human oversight, train clinicians on AI limitations, and ensure equitable infrastructure. Continuous auditing and performance monitoring across populations help sustain fairness in real-world use.
Data set shift	Occurs when data distributions change between training and deployment, reducing model accuracy and fairness. Particularly common in evolving clinical settings.	Temporal shift: changes in disease incidence or practice guidelines. Prevalence shift: different prevalence of risk factors or outcomes. Event-driven shift: arising from crises or policy changes (eg, COVID-19). Workflow shift: modifications in data generation processes.	A model predicting LVO prevalence performs well on neurologist-selected patients but fails in emergency cohorts. ⁵⁰ COVID-19 altered stroke presentation timing, reducing diagnostic sensitivity. ⁴⁹ Models trained on specific CT scanners underperform when deployed on new imaging protocols. ¹⁹	Adaptive postprocessing: Monitor and recalibrate models periodically. Apply domain adaptation, transfer learning, or threshold recalibration to maintain performance. When an annotation shift exists (eg, systematic underdiagnosis), reannotation by unbiased experts is crucial to correct the ground truth.

AI indicates artificial intelligence; and CT, computed tomography.

concerns regarding patient consent and data privacy.⁵⁷ One of the most pressing ethical concerns is ensuring that patients fully understand the scope of data being collected, how it will be used, and the potential risks to their privacy. Consent processes must be clear, comprehensive, and ongoing. Patients should be informed about the types of data collected and their potential uses, the duration for which data will be stored, and monitored the potential third-party access to their data. And the ability to withdraw consent at any point without jeopardizing the care they receive. Informed consent is not just a 1-time agreement; rather, it should be a continuous process. As DHTs evolve, patients must be periodically reeducated and given the opportunity to update their consent preferences.

Data Storage and Protection

Although DHTs greatly enhance stroke care, they also introduce significant risks related to data privacy and security. Patients with stroke health data, often stored in electronic databases, is vulnerable to cyberattacks, data breaches, unauthorized access, and ransomware.⁵⁸ Given the sensitive nature of medical information, particularly

from smart devices or digital tools, protecting this data is critical.⁵⁹ To mitigate these risks, health care providers and technology companies must: implement strong encryption protocols for secure data storage and transmission; conduct regular security audits and apply strict access controls to ensure that only authorized personnel can access patient information³⁴; prioritize patient informed consent and address the ethical and legal implications of data use, particularly in the context of digital vulnerability accordingly with the ethics committee's guidelines and ensure data integrity, as any accidental or malicious alteration could compromise patient safety and treatment outcomes.

Effective data protection is essential throughout the entire lifecycle of patient information to maintain trust and uphold the quality of care.

The Role of Deidentification Techniques

Deidentification is 1 potential means for facilitating the use of personally identifiable information in a way that does not identify or otherwise compromise the privacy of an individual or a group of individuals.⁶⁰ In stroke AI applications, deidentification strategies are essential

for protecting patient privacy, especially when working with sensitive data such as medical images, clinical records, and demographic variables. Techniques like data masking, anonymization, pseudonymization, and generalization help remove personal identifiers, enabling the ethical and legal use of data.⁶¹ Studies have shown that while deidentification enhances compliance with data protection laws like the Health Insurance Portability and Accountability Act and General Data Protection Regulation, it can also impact the granularity and analytical value of clinical data if not carefully implemented.⁶²

Third-Party Access to Data

Another significant ethical concern in digital stroke care is third-party access to sensitive health data, raising questions about who exactly has access to patient information. Is it limited to health care providers, or does it also include technology companies, research institutions, and insurance providers? The core issue lies in whether patients are fully informed about all the parties accessing their data and whether those entities are using the data ethically. Although some may require access for legitimate medical or research purposes, ethical concerns emerge when data is shared without proper patient knowledge or consent—especially when used for nonmedical purposes like marketing or commercial gain.⁵⁹ Ensuring transparency is essential to prevent the misuse of health data. Patients must be clearly informed about who has access to their data, why it is being shared, and what safeguards are in place to protect it. Moreover, health care providers and technology companies must empower patients with the ability to restrict or opt out of specific types of data sharing, particularly when the data is used for purposes beyond their direct care.⁶³

Regulations and Compliance

Several regulatory frameworks govern the use of health data, particularly for stroke DHTs. Regulatory frameworks, such as the Health Insurance Portability and Accountability Act in the United States and the General Data Protection Regulation in the EU, aim to protect patient privacy, ensure data security, and enforce transparency in data processing. In this scenario, machine unlearning plays an important role. Machine unlearning refers to the process of removing the influence of specific data points from a trained machine learning model, without requiring full retraining. This capability is increasingly important for ensuring compliance with data privacy regulations, which grant individuals the right to have their data erased.⁶⁴ In addition to privacy legislation, DHTs are often subject to regulatory frameworks—such as the US FDA or the EU medical device regulation—to demonstrate clinical safety and efficacy.

Federated Learning Solutions

FL represents a promising and disruptive approach to developing secure, privacy-preserving, and equitable AI systems in health care. Unlike traditional centralized models, which require pooling sensitive patient data into a single repository, FL enables multiple institutions to collaboratively train a shared machine learning model without transferring raw data.⁶⁵ Each site trains the model locally on its own clinical data and sends only encrypted updates (eg, weights or gradients) to a central server. These are aggregated into a global model and shared back with clients. This iterative process improves performance while keeping sensitive data on-site, aligning with regulations like the Health Insurance Portability and Accountability Act and GDPR (Figure). A key advantage of this architecture is that it allows the use of large and diverse data sets distributed across institutions, which enhances the model's generalizability and helps mitigate data set bias. By leveraging a broader population base and clinical variability, FL supports the development of more robust and inclusive AI tools. Advanced techniques like secure aggregation, differential privacy, and homomorphic encryption further protect against patient reidentification from shared model parameters.²⁷ Despite its benefits, FL also faces significant challenges. Chief among them is the heterogeneity of data across medical centers: differences in patient populations, diagnostic protocols, data collection, and equipment can lead to nonindependent and identically distributed (non-IID) data. To address this, local preprocessing methods, such as SMOTE (The Synthetic Minority Over Sampling Technique) for class balancing, can be applied, and customized algorithms are used to synchronize local updates with the global model. In some implementations, strategies are introduced to ensure that under-resourced or smaller institutions can meaningfully participate in the learning process, regardless of data volume.⁶⁶

Global Compliance Challenges

Although deidentification and FL offer important tools for privacy-preserving AI development, vendors face additional challenges due to varying regulatory requirements across different countries and regions. Standards for what constitutes sufficient deidentification, acceptable risk thresholds for reidentification, and definitions of data anonymization differ significantly between different legal frameworks and other national health data regulations. Governments worldwide are introducing regulations to keep pace with this technology. These inconsistencies and the complex regulatory maze complicate the deployment of stroke AI systems across borders. Moreover, FL may still face legal scrutiny regarding metadata transmission, model update auditing, and central coordination. Navigating this fragmented regulatory landscape

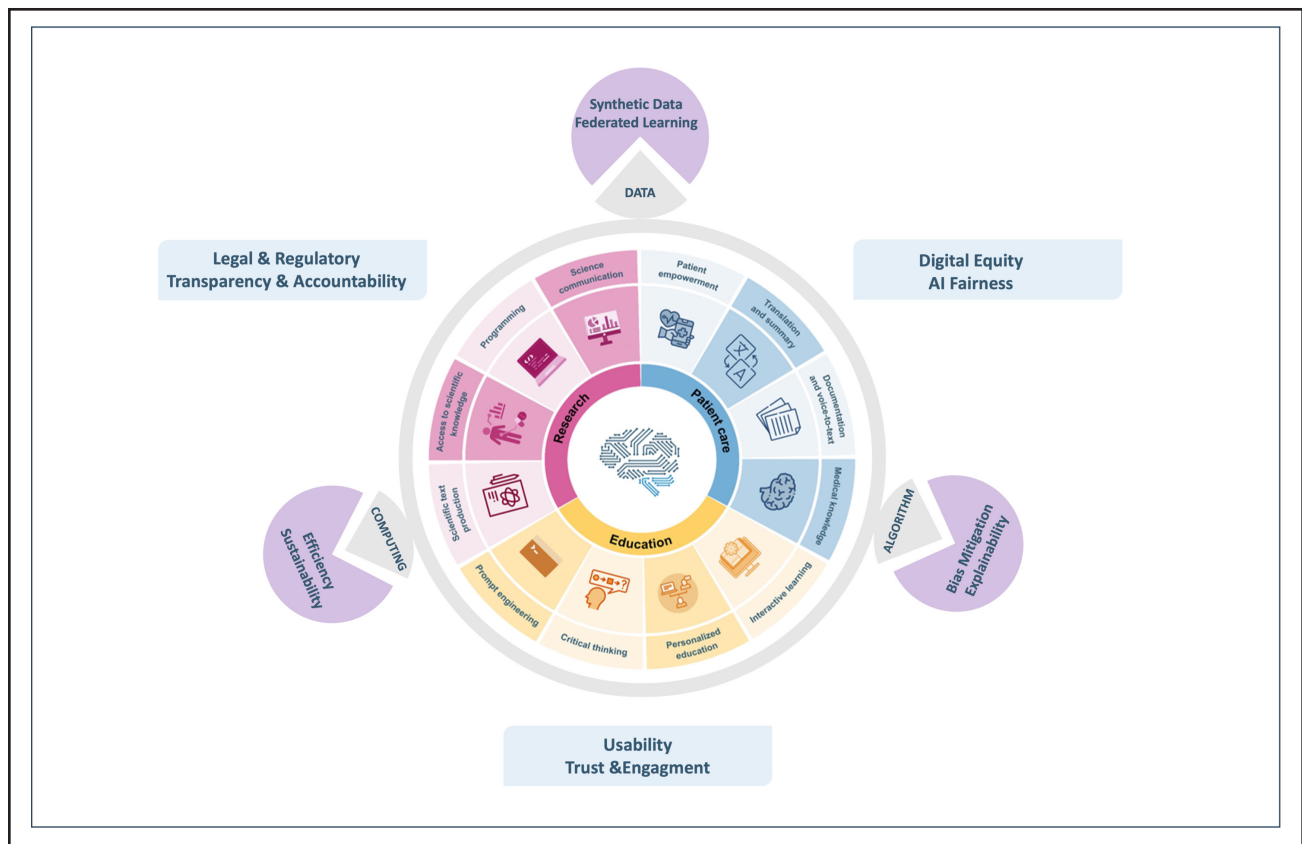


Figure. Conceptual framework summarizing the ethical and privacy challenges in stroke digital health.

The figure illustrates the multidimensional integration of digital technologies within stroke care, research, and education, emphasizing the interplay between ethical, regulatory, and usability principles. Central domains, education, research, and patient care, are surrounded by cross-cutting elements such as data governance, algorithmic fairness and bias mitigation, computing efficiency and sustainability, and user trust and engagement. Outer ethical pillars highlight digital equity, artificial intelligence (AI) fairness, transparency, accountability, legal regulation, and privacy as essential to ensure safe, equitable, and trustworthy digital health ecosystems in stroke management.

demands flexible, modular approaches to compliance and highlights the need for internationally harmonized standards in health data governance.⁶⁷

END-USER TRUST AND ENGAGEMENT

Beyond fulfilling ethical and privacy requirements, the effective integration of AI into clinical practice hinges on another key factor: trust. Trust is foundational to the adoption and sustained use of DHTs. It is particularly crucial in vulnerable populations such as stroke survivors, as well as their families and caregivers. For these individuals, confidence in technology must go hand in hand with a sense of safety, clarity, and support.⁶⁸ However, trust must not be limited to patients alone: clinicians also need to develop confidence in AI systems and digital solutions. This entails being able to rely on such tools to complement and enrich clinical information, with the shared objective of improving patient care.

Much of the trust in health care is historically rooted in human connection, through empathy, communication, and shared decision-making. These elements transform treatment into care. Unlike clinicians, AI systems do not

possess empathy or human intuition; they are tools, not partners. This distinction is important. It raises a central question: Can trust in AI be misplaced?⁶⁹ Perhaps it can, particularly if the entire system surrounding the development and continuous monitoring of AI fails to adhere to appropriate ethical and privacy frameworks. Without robust oversight, transparency, and accountability, trust in AI risks becoming superficial or even harmful, especially in clinical contexts where the stakes are high. This means that the nature and foundations of trust must shift. As AI becomes increasingly integrated into clinical care, we need to redefine what it means to trust, not by making AI more human, but by ensuring that it is transparent, accountable, and reliably suited for delegated tasks.⁷⁰ For patients with stroke specifically, trust in AI may be influenced by how well the technology integrates into their care plan, supports communication with providers, and contributes meaningfully to their recovery. AI must enhance the human capacity to care.

One of the most effective strategies for building trust is ensuring transparency. This includes clearly communicating how data is collected and used, how AI models make decisions, and how outcomes are generated.⁷¹ Patients

with stroke and caregivers are more likely to engage with digital tools when the technology is understandable and not perceived as a black box. Additionally, the physician plays a fundamental role; if they recommend a tool, that trust is likely to extend to the technology itself.

Concerns around whether DHTs might replace health care professionals often emerge in this debate. However, it is unlikely that AI will supplant clinicians. Rather, its role will be complementary, serving primarily as a support system for clinical decision-making and routine tasks.⁶⁹ AI is particularly well-suited for automating repetitive processes such as data collection, real-time monitoring, and preliminary image analysis, allowing health care professionals to focus on more complex, patient-centred care. Essential human qualities such as empathy and clinical judgment will remain crucial. Rather than replacing clinicians, AI will serve as a collaborative tool to enhance efficiency, diagnostic accuracy, and personalized treatment.

REGULATORY AND LEGAL PERSPECTIVES

The integration of DHTs in stroke care—such as AI diagnostics, telemedicine, mobile health applications, and wearable monitoring devices—offers significant clinical advantages. However, these innovations also raise complex legal issues, particularly around data privacy, consent, algorithmic fairness, and liability. This brief outlines the current regulatory and legal landscape, identifies key challenges, and offers policy recommendations to ensure the safe, ethical, and equitable deployment of digital tools in stroke care.⁶⁸ The regulatory approach to AI must be risk-based and resilient. A focus on unknown emerging risks demands continuous surveillance and clinical evaluation during the total product life cycle. Since learning algorithms are data-driven, high-quality data is fundamental for good machine learning practice. Mining, processing, validation, governance, and data control must account for bias, error, inappropriate use, drifts, and shifts, particularly in real-world data. Regulators worldwide are tackling 21st-century challenges raised by learning medical devices.

CONCLUSIONS

As AI becomes increasingly integrated into health care, a wide range of challenges arises, spanning from technological advancements to regulatory issues and ethical concerns. The rise of DHTs in stroke care has transformed the entire continuum of care, from prehospital triage and acute treatment to rehabilitation and long-term follow-up. The goal is not to replace clinicians with AI systems, but rather to enhance and support the capabilities of health care professionals, strengthening the collaborative work of clinical teams as they strive to provide more efficient, accurate, and patient-centred care.

Although these technologies promise to improve access, efficiency, and outcomes, they also pose significant ethical and legal challenges. It is essential to address a number of ethical concerns, such as data privacy and security, ensuring adequate informed consent, preventing algorithmic biases that could worsen health disparities, assigning responsibility for automated decisions, and preserving patient trust. Transparency and fairness are key elements in preventing discrimination and ensuring safe and equitable access to AI-powered health care. For effective and comprehensive adoption, AI systems must obtain appropriate regulatory approval, establish standardized protocols to guarantee consistent functionality, include proper training for clinicians and patients, ensure financial support, promote equity in implementation, and undergo continuous, iterative, and rigorous oversight. Addressing these challenges requires strong regulatory frameworks, ongoing supervision, and the active and coordinated participation of all stakeholders involved in the health care ecosystem. In this context, the engagement of patients and caregivers and their early involvement in the design of digital solutions are valuable and essential. Their lived experiences, preferences, and concerns provide critical insights that can guide the development, implementation, and evaluation of DHTs. Furthermore, active and early contribution of legal and ethics experts is crucial to ensure that AI solutions are trustworthy and ethical by design, complying with ethical standards and legal requirements. These stakeholders play a key role in safeguarding patients' rights, promoting transparency, and establishing accountability mechanisms, particularly as AI systems take on more influential roles in clinical decision-making.

By integrating the voices of patients, caregivers, legal professionals, and ethics bodies, health care systems can foster greater trust, legitimacy, and acceptance of AI-powered solutions. This inclusive and participatory approach not only enhances ethical compliance but also helps align technological innovation with the principles of patient-centred care, social justice, and public interest.

ARTICLE INFORMATION

Affiliations

Stroke Unit, Department of Neurology, Vall d'Hebron Research Institute, Vall d'Hebron University Hospital, Barcelona, Spain (M.M., F.R., N.B., C.A.M.). Department of Medicine, Universitat Autònoma de Barcelona, Spain (F.R.).

Sources of Funding

None.

Disclosures

Dr Molina is the principal investigator of 2 projects funded by the European Commission: TRUSTroke (Trustworthy Prediction of Stroke Outcome on a Federated Learning Infrastructure; TRUSTROKE_HE-HLTH-STAYHLTH2022) and UMBRELLA (Unleashing a Comprehensive, Holistic and Patient Centric Stroke Management for a Better, Rapid, Advanced and Personalised Stroke Diagnosis, Treatment and Outcome Prediction; UMBRELLA_HE_IHI2024).

The other authors report no conflicts.

REFERENCES

- Kaihanen AM, Virtanen L, Buchert U, Safarov N, Valkonen P, Hietapakka L, Hörhammer I, Kujala S, Kouvonen A, Jormanainen V, et al. Towards digital health equity - a qualitative study of the challenges experienced by vulnerable groups in using digital health services in the COVID-19 era. *BMC Health Serv Res*. 2022;22:188. doi: 10.1186/s12913-022-07584-4
- Verma A, Towfighi A, Brown A, Abhat A, Casillas A. Moving towards equity with digital health innovations for stroke care. *Stroke*. 2022;53:689–697. doi: 10.1161/STROKEAHA.121.035307
- Richardson S, Lawrence K, Schoenthaler AM, Mann D. A framework for digital health equity. *NPJ Digit Med*. 2022;5:119. doi: 10.1038/s41746-022-00663-0
- Grande D, Luna Marti X, Feuerstein-Simon R, Merchant RM, Asch DA, Lewson A, Cannuscio CC. Health policy and privacy challenges associated with digital health technology. *JAMA Netw Open*. 2020;3:e208285. doi: 10.1001/jamanetworkopen.2020.8285
- Jauch EC, Schwamm LH, Panagos PD, Barbazzani J, Dickson R, Dunne R, Foley J, Fraser JF, Lassers G, Martin-Gill C, et al; Prehospital Stroke System of Care Consensus Conference. Recommendations for regional stroke destination plans in rural, suburban, and urban communities from the prehospital stroke system of care consensus conference: a consensus statement from the American Academy of Neurology, American Heart Association/American Stroke Association, American Society of Neuroradiology, National Association of EMS Physicians, National Association of State EMS Officials, Society of NeuroInterventional Surgery, and Society of Vascular and Interventional Neurology: Endorsed by the Neurocritical Care Society. *Stroke*. 2021;52:e133–e152. doi: 10.1161/STROKEAHA.120.033228
- Nguyen TTM, Van Den Wijngaard IR, Bosch J, Boiten J, Wermer MJH, Groot AE, Algra A, Kruijff ND, Lingsma HF, Dippel DWJ, et al. Comparison of prehospital scales for predicting large anterior vessel occlusion in the ambulance setting. *JAMA Neurol*. 2021;78:157. doi: 10.1001/jamaneurol.2020.4418
- Fatima N, Saqqur M, Hussain MS, Shuaib A. Mobile stroke unit versus standard medical care in the management of patients with acute stroke: a systematic review and meta-analysis. *Int J Stroke*. 2020;15:595–608. doi: 10.1177/1747493020929964
- English SW, Barrett KM, Freeman WD, Demaerschalk BM. Telemedicine-enabled ambulances and mobile stroke units for prehospital stroke management. *J Telemed Telecare*. 2022;28:458–463. doi: 10.1177/1357633X211047744
- Bat-Erdene BO, Saver JL. Automatic acute stroke symptom detection and emergency medical systems alerting by mobile health technologies: a review. *J Stroke Cerebrovasc Dis*. 2021;30:105826. doi: 10.1016/j.jstrokecerebrovasdis.2021.105826
- Wasselius J, Lyckegård Finn E, Persson E, Ullberg T, Hansen T, Cronqvist M, Wassélius M, Söderman M, Holmin S, Gällstedt M, et al. Detection of unilateral arm paresis after stroke by wearable accelerometers and machine learning. *Sensors*. 2021;21:7784. doi: 10.3390/s21237784
- Kim H, Kim S, Kim Y, Park K. A smartphone-based automatic diagnosis system for facial nerve palsy. *Sensors*. 2015;15:26756–26768. doi: 10.3390/s151026756
- English SW, Barrett KM, Freeman WD, Demaerschalk BM, Dumitrascu OM. Improving the telemedicine evaluation of patients with acute vision loss: a call to eyes. *Neurology*. 2022;99:381–386. doi: 10.1212/WNL.000000000000200969
- Scholz ML, Collatz-Christensen H, Blomberg SNF, Boebel S, Verhoeven J, Krafft T. Artificial intelligence in Emergency Medical Services dispatching: assessing the potential impact of an automatic speech recognition software on stroke detection taking the Capital Region of Denmark as case in point. *Scand J Trauma Resusc Emerg Med*. 2022;30:36. doi: 10.1186/s13049-022-01020-6
- Oostema JA, Chassee T, Reeves M. Emergency dispatcher stroke recognition: associations with downstream care. *Prehosp Emerg Care*. 2018;22:466–471. doi: 10.1080/10903127.2017.1405131
- Ceklic E, Ball S, Finn J, Cox S, Ackerly J, Anderson J, Burns B, Smith K. Ambulance dispatch prioritisation for traffic crashes using machine learning: a natural language approach. *Int J Med Inf*. 2022;168:104886. doi: 10.1016/j.ijmedinf.2022.104886
- Schwamm LH. Digital health strategies to improve care and continuity within stroke systems of care in the United States. *Circulation*. 2019;139:149–151. doi: 10.1161/CIRCULATIONAHA.117.029234
- Soun JE, Chow DS, Nagamine M, Takhtawala RS, Filippi CG, Yu W, Chang PD. Artificial intelligence and acute stroke imaging. *AJNR Am J Neuroradiol*. 2021;42:2–11. doi: 10.3174/ajnr.A6883
- Mouridsen K, Thurner P, Zaharchuk G. Artificial intelligence applications in stroke. *Stroke*. 2020;51:2573–2579. doi: 10.1161/STROKEAHA.119.027479
- Parvathy G, Kamaraj B, Sah B, Konda M, Dave T, Kanwar P, Roy S, Deora H. Emerging artificial intelligence-aided diagnosis and management methods for ischemic strokes and vascular occlusions: a comprehensive review. *World Neurosurg*. 2024;22:100303. doi: 10.1016/j.wnsx.2024.100303
- Song X, Wang J, He F, Yin W, Ma W, Wu J. Stroke diagnosis and prediction tool using ChatGLM: development and validation study. *J Med Internet Res*. 2025;27:e67010. doi: 10.2196/67010
- Rubiera M, Aires A, Antonenko K, Bachus E, Fischer U, Gumbinger C, Hoyer C, Kallmünzer B, Katan Kahles M, Kellett J, et al. European Stroke Organisation (ESO) guideline on screening for subclinical atrial fibrillation after stroke or transient ischaemic attack of undetermined origin. *Eur Stroke J*. 2022;7:CVII–CXXXIX. doi: 10.1177/23969873221099478
- Sanna T, Diener HC, Passman RS, Di Lazzaro V, Bernstein RA, Morillo CA, Rymer MM, Thijs V, Rogers T, Beckers F, et al; CRYSTAL AF Investigators. Cryptogenic stroke and underlying atrial fibrillation. *N Engl J Med*. 2014;370:2478–2486. doi: 10.1056/NEJMoa1313600
- Bernstein RA, Kamel H, Granger CB, Piccini JP, Sethi PP, Katz JM, Vives CA, Schwamm LH, Bernstein RA, Elkind MSV, et al. Atrial fibrillation in patients with stroke attributed to large- or small-vessel disease: 3-year results from the STROKE AF randomized clinical trial. *JAMA Neurol*. 2023;80:1277. doi: 10.1001/jamaneurol.2023.3931
- Gladstone DJ, Spring M, Dorian P, Panzov V, Thorpe KE, Hall J, Vaid H, O'Donnell M, Laupacis A, Côté R, et al; EMBRACE Investigators and Coordinators. Atrial fibrillation in patients with cryptogenic stroke. *N Engl J Med*. 2014;370:2467–2477. doi: 10.1056/NEJMoa1311376
- Meza C, Juega J, Francisco J, Soteras J, Terceno M, Remollo S, Werner M, Becerra JL, Vera M, Torné R, et al. Accuracy of a smartwatch to assess heart rate monitoring and atrial fibrillation in stroke patients. *Sensors*. 2023;23:4632. doi: 10.3390/s23104632
- Cramer SC, Dodakian L, Le V, See J, Augsburg R, McKenzie A, Zhou RJ, Chiu NL, Heckhausen J, Cassidy JM, et al; National Institutes of Health StrokeNet Telerehab Investigators. Efficacy of home-based telerehabilitation vs in-clinic therapy for adults after stroke: a randomized clinical trial. *JAMA Neurol*. 2019;76:1079–1087. doi: 10.1001/jamaneurol.2019.1604
- Chen RJ, Wang JJ, Williamson DFK, Zhao S, Lipkova J, Lu MY, Sahai S, Mahmood F. Algorithmic fairness in artificial intelligence for medicine and healthcare. *Nat Biomed Eng*. 2023;7:719–742. doi: 10.1038/s41551-023-01056-8
- Requena M, Montiel E, Baladas M, Muchada M, Boned S, López R, Rodríguez M, Juega J, Pagola J, Molina C, et al. Farmalarm: application for mobile devices improves risk factor control after stroke. *Stroke*. 2019;50:1819–1824. doi: 10.1161/STROKEAHA.118.024355
- Cao W, Kadir AA, Tang W, Wang J, Yuan J, Hassan II. Effectiveness of mobile application interventions for stroke survivors: systematic review and meta-analysis. *BMC Med Inform Decis Mak*. 2024;24:6. doi: 10.1186/s12911-023-02391-1
- Olive-Gadea M, Cano D, Rodrigo-Gisbert M, Requena M, Muchada M, Rubiera M, Boned S, Tomasello A, Pagola J, Juega J, et al. Redefining disability: patient-reported outcome measures after minor stroke and transient ischemic attack. *Stroke*. 2023;54:144–150. doi: 10.1161/STROKEAHA.122.040409
- Allan LP, Silvera-Tawil D, Cameron J, Donnelly A, Gorman C, Hemsley B, Hsu HY, Jones K, Kohler F, Nguyen K, et al. Novel multicomponent digital care assistant and support program for people after stroke or transient ischaemic attack: a pilot feasibility study. *Sensors*. 2024;24:7253. doi: 10.3390/s24227253
- Samak ZA, Clatworthy P, Mirmehdi M. Automatic prediction of stroke treatment outcomes: latest advances and perspectives. *Biomed Eng Lett*. 2025;15:467–488. doi: 10.1007/s13534-025-00462-y
- García-Terriza L, Risco-Martín JL, Roselló GR, Ayala JL. Predictive and diagnosis models of stroke from hemodynamic signal monitoring. *Med Biol Eng Comput*. 2021;59:1325–1337. doi: 10.1007/s11517-021-02354-6
- Liu NF, Zhang T, Liang P. Evaluating verifiability in generative search engines. In: *Findings of the Association for Computational Linguistics: EMNLP*. 2023;7001–7025. doi: 10.18653/v1/2023.findings-emnlp.467
- Colangelo G, Ribo M, Montiel E, Baladas M, Muchada M, Rodríguez M, Juega J, Pagola J, Molina C, Tomasello A, et al. PRERISK: a personalized, artificial intelligence-based and statistically-based stroke recurrence predictor for recurrent stroke. *Stroke*. 2024;55:1200–1209. doi: 10.1161/STROKEAHA.123.043691

36. Tunkl C, Agarwal A, Ramage E, Avila-Vargas P, Hidalgo K, Sacks C, Switzer JA, Demaerschalk BM, Golanov EV, Zerna C, et al. Telemedicine networks for acute stroke: an analysis of global coverage, gaps, and opportunities. *Int J Stroke*. 2025;20:297–309. doi: 10.1177/17474930241298450
37. Bitomsky L, Pfitzer E, Nißen M, Kowatsch T. Advancing health equity and the role of digital health technologies: a scoping review. *BMJ Open*. 2025;15:e099306. doi: 10.1136/bmjopen-2025-099306
38. Adler NE, Glymour MM, Fielding J. Addressing social determinants of health and health inequalities. *JAMA*. 2016;316:1641–1642. doi: 10.1001/jama.2016.14058
39. Nouri S, Khoong EC, Lyles CR, Karliner L. Addressing equity in telemedicine for chronic disease management during the COVID-19 pandemic. Published online 2020.
40. Ratwani RM, Sutton K, Galarraga JE. Addressing AI algorithmic bias in health care. *JAMA*. 2024;332:1051–1052. doi: 10.1001/jama.2024.13486
41. Hong C, Pencina MJ, Wojdyla DM, Hall JL, Judd SE, Cary M, Engelhard MM, Xian Y, Alger HM, Navar AM, et al. Predictive accuracy of stroke risk prediction models across black and white race, sex, and age groups. *JAMA*. 2023;329:306. doi: 10.1001/jama.2022.24683
42. Li DM, Parikh S, Costa A. A critical look into artificial intelligence and healthcare disparities. *Front Artif Intell*. 2025;8:1545869. doi: 10.3389/frai.2025.1545869
43. Finlayson SG, Subbaswamy A, Singh K, Bowers J, Kupke A, Zittrain J, Kohane IS, Saria S. The clinician and dataset shift in artificial intelligence. *N Engl J Med*. 2021;385:283–286. doi: 10.1056/NEJMc2104626
44. Wiens J, Saria S, Sendak M, Ghassemi M, Liu VX, Doshi-Velez F, Jung K, Heller K, Kale D, Saeed M, et al. Do no harm: a roadmap for responsible machine learning for health care. *Nat Med*. 2019;25:1337–1340. doi: 10.1038/s41591-019-0548-6
45. Chen F, Wang L, Hong J, Jiang J, Zhou L. Unmasking bias in artificial intelligence: a systematic review of bias detection and mitigation strategies in electronic health record-based models. *J Am Med Inform Assoc*. 2024;31:1172–1183. doi: 10.1093/jamia/ocae060
46. Nazer LH, Zatarah R, Waldrip S, Ke JXC, Moukheiber M, Khanna AK, Hicklen RS, Moukheiber L, Moukheiber D, Ma H, et al. Bias in artificial intelligence algorithms and recommendations for mitigation. *PLOS Digit Health*. 2023;2:e0000278. doi: 10.1371/journal.pdig.0000278
47. Parmar P, Krishnamurthi R, Ikram MA, Hofman A, Mirza SS, Varakin Y, Kravchenko M, Piradov M, Thrift AG, Norrving B, et al; Stroke RiskometerTM Collaboration Writing Group. The Stroke RiskometerTM App: validation of a data collection tool and stroke risk predictor. *Int J Stroke*. 2015;10:231–244. doi: 10.1111/ijis.12411
48. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*. 2019;366:447–453. doi: 10.1126/science.aax2342
49. Leslie D, Mazumder A, Peppin A, Wolters MK, Hagerty A. Does “AI” stand for augmenting inequality in the era of COVID-19 healthcare? *BMJ*. 2021;372:n304. doi: 10.1136/bmj.n304
50. Olivé-Gadea M, Mayol J, Requena M, Muchada M, Rubiera M, Boned S, Tomasello A, Pagola J, Juega J, Rodriguez M, et al. Impact of pre-test probability on AI-LVO detection: a systematic review of LVO prevalence across clinical contexts. *J NeuroIntervent Surg*. 2025. doi: 10.1136/jnis-2025-023775
51. Rajkomar A, Hardt M, Howell MD, Corrado G, Chin MH. Ensuring fairness in machine learning to advance health equity. *Ann Intern Med*. 2018;169:866–872. doi: 10.7326/M18-1990
52. Doshi-Velez F, Kim B. Towards a rigorous science of interpretable machine learning. *arXiv*. 2017. doi: 10.48550/arXiv.1702.08608
53. Matheny ME, Goldsack JC, Saria S, Corrigan-Curay J, Pencina M, Tenaerts P, Weinfurt K. Artificial intelligence in health and health care: priorities for action: article examines priorities for the uses of artificial intelligence in health and health care. *Health Aff (Millwood)*. 2025;44:163–170. doi: 10.1377/hlthaff.2024.01003
54. WHO Expert Group on Ethics and Governance of AI for Health. *Ethics and governance of artificial intelligence for health: WHO guidance*. World Health Organization; 2021.
55. Conduah AK, Ofuo S, Siaw-Marfo D. Data privacy in healthcare: global challenges and solutions. *Digit Health*. 2025;11:20552076251343959. doi: 10.1177/20552076251343959
56. Grossmann C, Goolsby A, Olsen L, McGinnis JM. *Clinical data as the basic staple of health learning: creating and protecting a public good: workshop summary*. National Academies Press; 2010. doi:10.17226/12212
57. Kassam I, Ilkina D, Kemp J, Roble H, Carter-Langford A, Shen N. Patient perspectives and preferences for consent in the digital health context: state-of-the-art literature review. *J Med Internet Res*. 2023;25:e42507. doi: 10.2196/42507
58. Torgersen LNS, Schulz SM, Lugo RG, Sütterlin S. Patient informed consent, ethical and legal considerations in the context of digital vulnerability with smart, cardiac implantable electronic devices. *PLOS Digit Health*. 2024;3:e0000507. doi: 10.1371/journal.pdig.0000507
59. Mittelstadt BD, Allo P, Taddeo M, Wachter S, Floridi L. The ethics of algorithms: Mapping the debate. *Big Data Soc*. 2016;3:2053951716679679. doi: 10.1177/2053951716679679
60. Kondylakis H, Catalan R, Alabart SM, Bitterman DS, Buvat I, Carot Sierra JM, Frangi AF, Fuller CD, Neri E, Prior F, et al. Documenting the de-identification process of clinical and imaging data for AI for health imaging projects. *Insights Imaging*. 2024;15:130. doi: 10.1186/s13244-024-01711-x
61. Langarizadeh M, Orooji A, Sheikhtaheri A. Effectiveness of anonymization methods in preserving patients' privacy: a systematic literature review. *Stud Health Technol Inform*. 2018;248:80–87. doi: 10.3233/978-1-61499-858-7-80
62. Rothstein MA. Is deidentification sufficient to protect health privacy in research? *Am J Bioeth*. 2010;10:3–11. doi: 10.1080/15265161.2010.494215
63. Vayena E, Blasimme A. Health research with big data: time for systemic oversight. *J Law Med Ethics*. 2018;46:119–129. doi: 10.1177/1073110518766026
64. Zhou J, Li H, Liao X, Wang S, Trappey A, Zhang JJ, Gao J, Sun M. A unified method to revoke the private data of patients in intelligent healthcare with audit to forget. *Nat Commun*. 2023;14:6255. doi: 10.1038/s41467-023-41703-x
65. Abbas SR, Abbas Z, Zahir A, Lee SW. Federated learning in smart healthcare: a comprehensive review on privacy, security, and predictive analytics with IoT integration. *Healthcare (Basel)*. 2024;12:2587. doi: 10.3390/healthcare12242587
66. Sowjanya AM, Mrudula O. Effective treatment of imbalanced datasets in health care using modified SMOTE coupled with stacked deep learning algorithms. *Appl Nanosci*. 2023;13:1829–1840. doi: 10.1007/s13204-021-02063-4
67. Schwalbe N, Wahl B. Artificial intelligence and the future of global health. *Lancet*. 2020;395:1579–1586. doi: 10.1016/S0140-6736(20)30226-9
68. Mennella C, Maniscalco U, De Pietro G, Esposito M. Ethical and regulatory challenges of AI technologies in healthcare: a narrative review. *Heliyon*. 2024;10:e26297. doi: 10.1016/j.heliyon.2024.e26297
69. Teo ZL, Jin L, Liu N, Chandrasekaran V, Chen Y, Ting DSW. Federated machine learning in healthcare: a systematic review on clinical applications and technical architecture. *Cell Rep Med*. 2024;5:101419. doi: 10.1016/j.xcrm.2024.101419
70. Beger J. Not someone, but something: rethinking trust in the age of medical AI. *Eur J Radiol AI*. 2025;3:100038. doi: 10.1016/j.ejrai.2025.100038
71. Zhan X, Abdi N, Seymour W, Such J. Healthcare voice AI assistants: factors influencing trust and intention to use. *Proc ACM Hum-Comput Interact*. 2024;8:1–37. doi: 10.1145/3637339